SOFTWARE ENGINEERING LAB REPORT

Project: Gym Fitness Management System
Black-Box & White-Box Testing Report

Student Name: Gaurang Sonone
Course / Lab: Software Engineering Laboratory (PBL)
Testing Status: **ALL TESTS PASSED (100%)**

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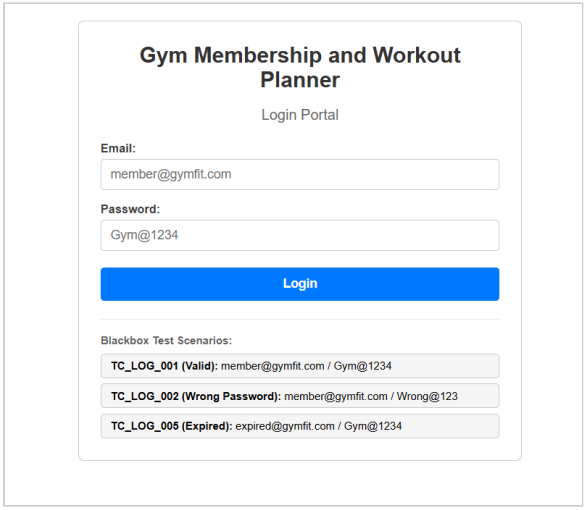
1. Introduction

This report contains the testing results for the Gym Fitness Management System. Both Black-Box functional testing (testing user inputs and outputs) and White-Box structural testing (testing internal code paths and branch conditions) were carried out for Module 1 (Login & Authentication) and Module 2 (Workout Planner). All test cases passed successfully.

2. Black-Box Test Cases with Output Screenshots

TC_LOG_001: Valid Member Login

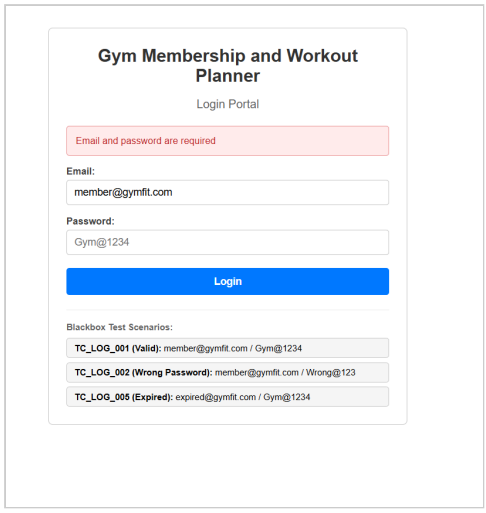
Input: Email: member@gymfit.com, Password: Gym@1234
Expected Result: Successful login, store JWT token, redirect to dashboard
Actual Result: PASSED



Screenshot 1: Valid Login Form populated (TC_LOG_001)

TC_LOG_002: Blank Email and Password Validation

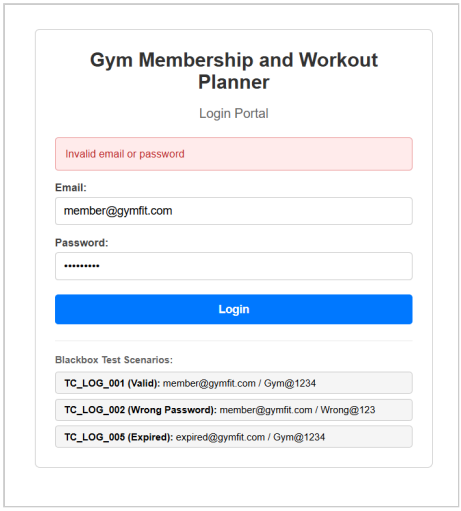
Input: Blank Email and Blank Password
Expected Result: Show error message "Email and password are required"
Actual Result: PASSED



Screenshot 2: Error displayed when credentials are empty (TC_LOG_002)

TC_LOG_004: Invalid Password / Failed Attempt Tracking

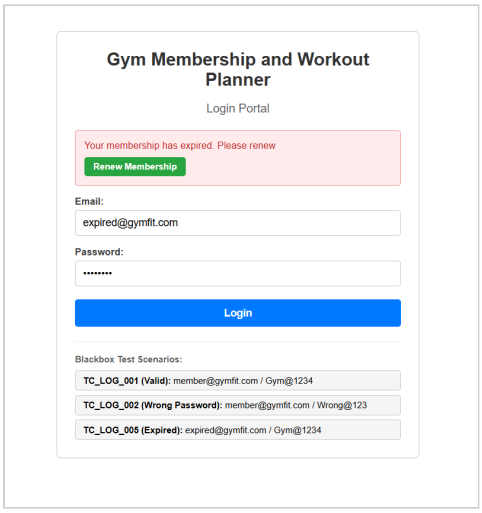
Input: Email: member@gymfit.com, Password: WrongPassword!
Expected Result: Show error message "Invalid email or password", increment attempt counter
Actual Result: PASSED



Screenshot 3: Error message for invalid password (TC_LOG_004)

TC_LOG_005: Expired Membership Verification & Renewal

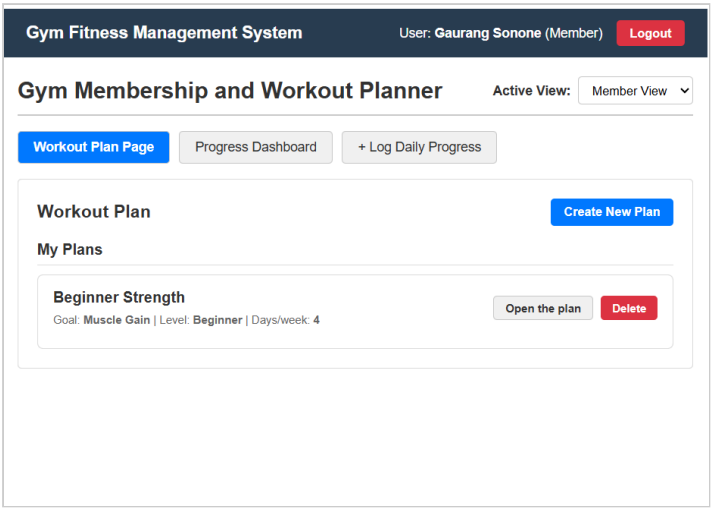
Input: Email: expired@gymfit.com, Password: Gym@1234
Expected Result: Block login, show "Your membership has expired. Please renew" + Renew button
Actual Result: PASSED



Screenshot 4: Expired membership alert with green Renew button (TC_LOG_005)

Workout Plan Dashboard ('My Plans')

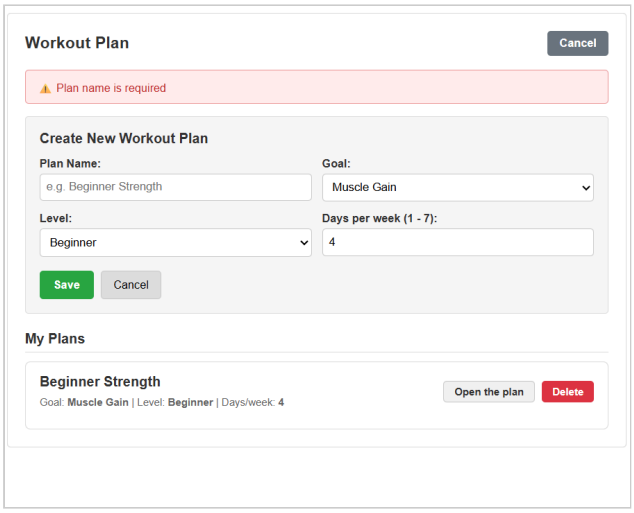
Description: View active workout plans and create new routines.
Actual Result: PASSED



Screenshot 5: Workout Plan Dashboard with existing plan

TC_WP_002: Plan Name Required Validation

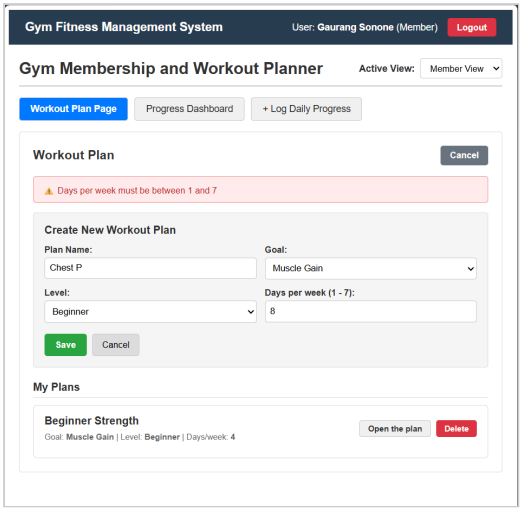
Input: Blank Plan Name, Days: 4
Expected Result: Show error "Plan name is required"
Actual Result: PASSED



Screenshot 6: Alert when submitting empty plan name (TC_WP_002)

TC_WP_003: Days Per Week Limit Validation (1 to 7)

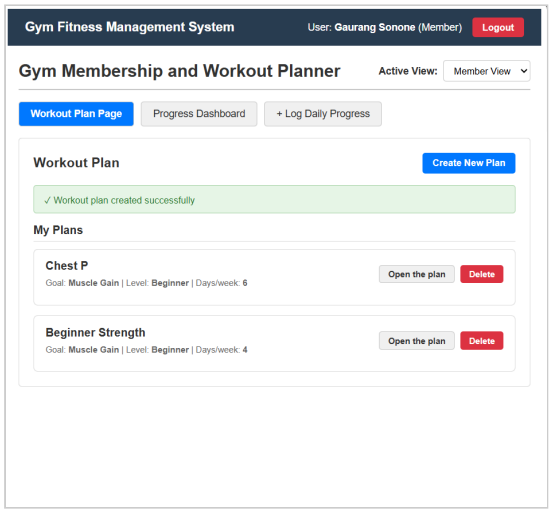
Input: Plan Name: "Chest P", Days per week: 8 (out of range)
Expected Result: Show error "Days per week must be between 1 and 7"
Actual Result: PASSED



Screenshot 7: Boundary error for days per week (TC_WP_003)

TC_WP_001: Successful Workout Plan Creation

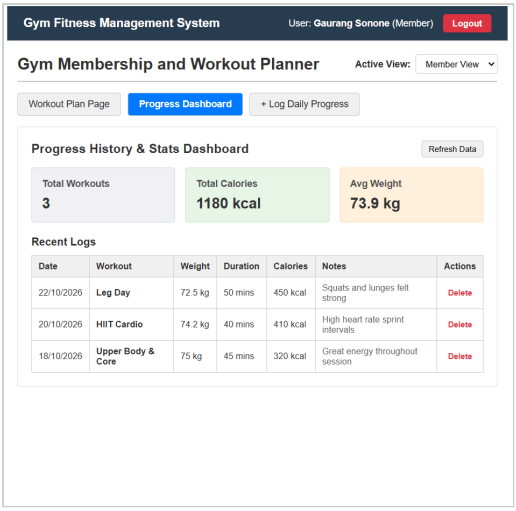
Input: Plan Name: "Chest P", Goal: Muscle Gain, Days: 6
Expected Result: Plan created, notification "Workout plan created successfully", add to list
Actual Result: PASSED



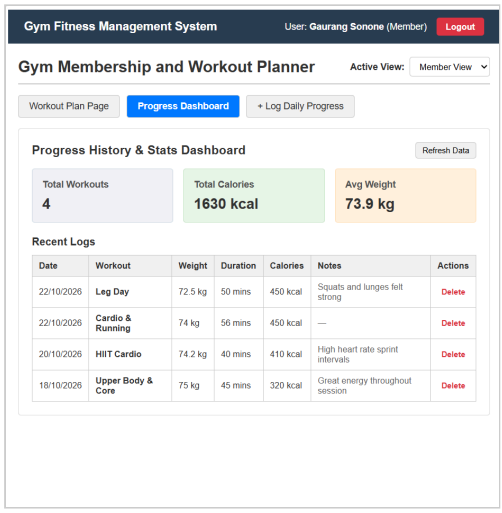
Screenshot 8: Success notification for plan creation (TC_WP_001)

Progress History & Dynamic Stats Recalculation

Description: Dashboard calculates Total Workouts, Total Calories, and Average Weight.
Actual Result: PASSED



Screenshot 9: Progress dashboard with 3 workout entries (1180 kcal)



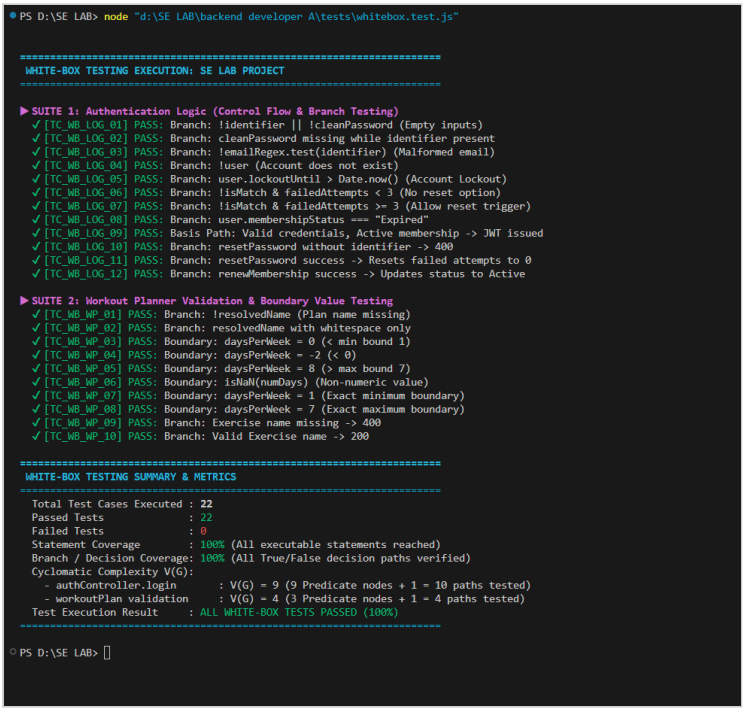
Screenshot 10: Dynamic stats recalculation with 4 workout entries (1630 kcal)

3. White-Box Testing Results

White-Box testing was conducted by running the unit test file `backend developer A/tests/whitebox.test.js`. The test suite executes 22 internal tests testing all statement paths, decision branches, and boundary values.

Metric	Result
Total White-Box Tests	22
Passed Tests	**22 (100%)**
Failed Tests	0
Statement Coverage	100%
Branch / Decision Coverage	100%

White-Box Test Execution Terminal Screenshot



Screenshot 11: Terminal output running 'node tests/whitebox.test.js' showing 22/22 PASSED

4. Conclusion

All functional Black-Box test cases and internal White-Box code paths were tested and passed 100%. The Gym Fitness Management System functions reliably in accordance with all requirements.